

Question ID: 95ca2683

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question

In triangle ABC , the measure of angle A is 54° , the measure of angle B is 90° , and the measure of angle C is $\left(\frac{k}{2}\right)^\circ$. What is the value of k ?

Answer

- A. 36
- B. 45
- C. 72
- D. 108

Correct Answer: C

Rationale

Choice C is correct. The sum of the interior angles of a triangle is 180° . It's given that the interior angles of triangle ABC are 54° , 90° , and $\left(\frac{k}{2}\right)^\circ$. It follows that $54 + 90 + \frac{k}{2} = 180$, or $144 + \frac{k}{2} = 180$. Subtracting 144 from each side of this equation yields $\frac{k}{2} = 36$. Multiplying each side of this equation by 2 yields $k = 72$. Therefore, the value of k is 72.

Choice A is incorrect. This is the value of $\frac{k}{2}$, not k .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.